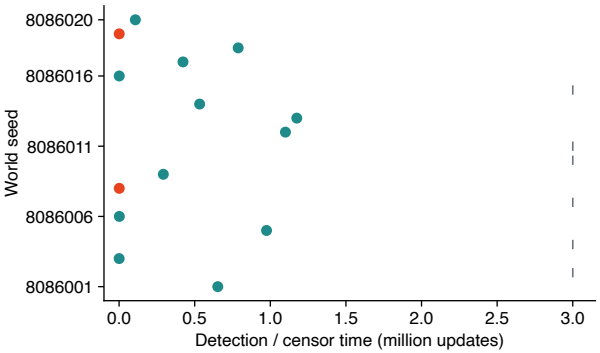
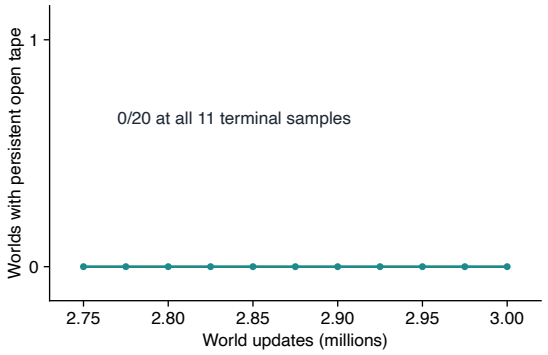
a **8086 S2: registered first-copier detection**



S2: coherent-fetch emergence; 20 worlds; L = 16 bytes
14/20 detected; 6/20 censored at 3 million updates
Teal dots: confined, 0/16 traces; red dots: open, ≥8/16 traces
Counted trace = any partner opcode/operand byte fetch
No first cases at 1–7/16; grey ticks = right censor

b **R2: descriptive open-copier persistence**



Same open fetch class as a; persistence also requires memory feeding
Same 20 S2 worlds; terminal persistence = 0/20
All earlier scheduled samples combined: 1/20 worlds ever positive

c **Registered evolved law: untested**

Family test	Candidate tapes	Eligible tapes	Unresolved tapes
L1	1337	0	0
L2	1337	0	724
L2 excluding L1	1337	0	724

L2 support suspended: zero certified cases

L1: frozen overlap; L2: broader pure-copy certificate

Zero eligible cases: law tests untested; capacity undetermined

c–e: candidate tapes from same 20 S2 worlds as a
L1 frozen overlap / L2 broader pure-copy certificate; d = L2 ledger

d **Registered exclusions and unresolved cases**

Status	Reason	Candidate tapes, n
culture-negative	culture not positive	558
excluded	original partner read	55
unresolved	Opcode 07 (POP ES)	708
unresolved	Opcode 0C	2
unresolved	Opcode 0F	1
unresolved	Opcode 21	1
unresolved	Opcode 31	1
unresolved	Opcode 5F	10
unresolved	Opcode E5	1

Total: 1,337 candidate tapes; 724 unresolved

L2 candidate tapes: 558 culture-negative, 55 excluded
724 unresolved (708 at POP ES); total 1,337

e **Post hoc: admissions and exact-law checks**

d bytes / edit	Genotypes	Worlds	Encounters with controls	Byte/core/ancestry mismatches
4 / erasure	10	10	538,240	0
16 / single/joint payload	22	20	1,686,400	0

d = write-minus-read displacement mod 32, in bytes
Paired address ring: donor A 0–15, partner B 16–31

Same 20 S2 worlds: 779 culture-positive world/tape records

Then 600 admitted records (590 d = 16; 10 d = 4); 32 selected genotypes

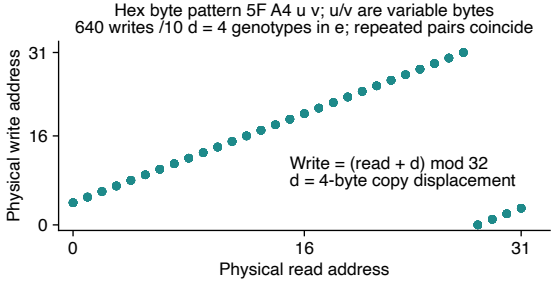
d = 4 joint free-payload tests: 0 applicable positions/genotype (n = 10)

d = 16 erasure tests: 0 applicable positions/genotype (n = 22)

32 genotypes /20 worlds; 2,224,640 encounters; worlds overlap by d

Observed offspring bytes, fixed core and ancestry match copy-law predictions
Nonempty perturbations and empty-assignment controls included in table
No applicable coordinates above means no corresponding perturbations
Listed encounters include empty-assignment controls; check bytes/core/ancestry

f **Exploratory byte-pattern copies: d = 4 bytes**



Separate snapshot fates: pooled median targets /512 encounters

Cohort	Median targets
8086 S2: 20 L16 worlds; 3 donor motifs	94.5
Z80 Stage G: 20 L16 worlds; 1 donor motif	214.5

Target: new non-donor 16-byte half matching a donor 2-byte tiling

Pools 5 snapshots/world × both soup and uniform partners × donor A/B
G = random-code cohort; motifs = distinct tiled 2-byte donor sequences